INDEPENDENT RESEARCH / NASDAQ-100 E-MINI FUTURES

NASDAQ Futures Quantitative Research Report

Daily trend-following strategy, transaction-cost model, chronological evaluation and data-quality audit.

EXECUTIVE FINDING

Positive simulated strategy P&L, but significant underperformance versus the always-long benchmark. The fixed 20/100-session moving-average rule produced $69,150 over the chronological 2023-2026 evaluation period. Holding one NQ contract long produced $335,540 on the same $250,000 reference capital and contract-multiplier basis. The strategy therefore lagged by $266,390.

Held-out metric (2023 - Nov 2026)	20/100 SMA	Always-long NQ
Net simulated P&L	$69,150	$335,540
Simple P&L / start capital	+27.66%	+134.22%
Daily-P&L Sharpe (0% benchmark)	0.20	0.96
Max model-equity drawdown	-29.52%	-21.75%
Annualised P&L volatility	36.28%	36.21%

Interpretation

The strategy did not meet the research hypothesis of improving risk-adjusted performance. It produced a much lower Sharpe ratio, a deeper drawdown, and substantially less net P&L than the comparison. These simulated dollar amounts are not live profits and are not a compounded investment return.

Study window and trading assumptions

The source spans 2019-01-02 to 2026-11-11 (1,980 daily bars). 2019-2022 is the research development window; 2023 onward is a chronological evaluation of fixed rules. Positions use the previous close signal, enter at the next recorded open, and accrue modelled open-to-next-open futures P&L. The model holds one $20-per-index-point NQ contract and charges $7.50 per execution side.

Evidence level: illustrative. The Yahoo-derived continuous NQ=F series has unverified roll adjustments, a source snapshot without a historical point-in-time audit, and no authenticated contract-chain execution path. Do not treat the figures as a genuinely tradable track record.

PERFORMANCE / CHRONOLOGICAL EVALUATION

Strategy versus benchmark

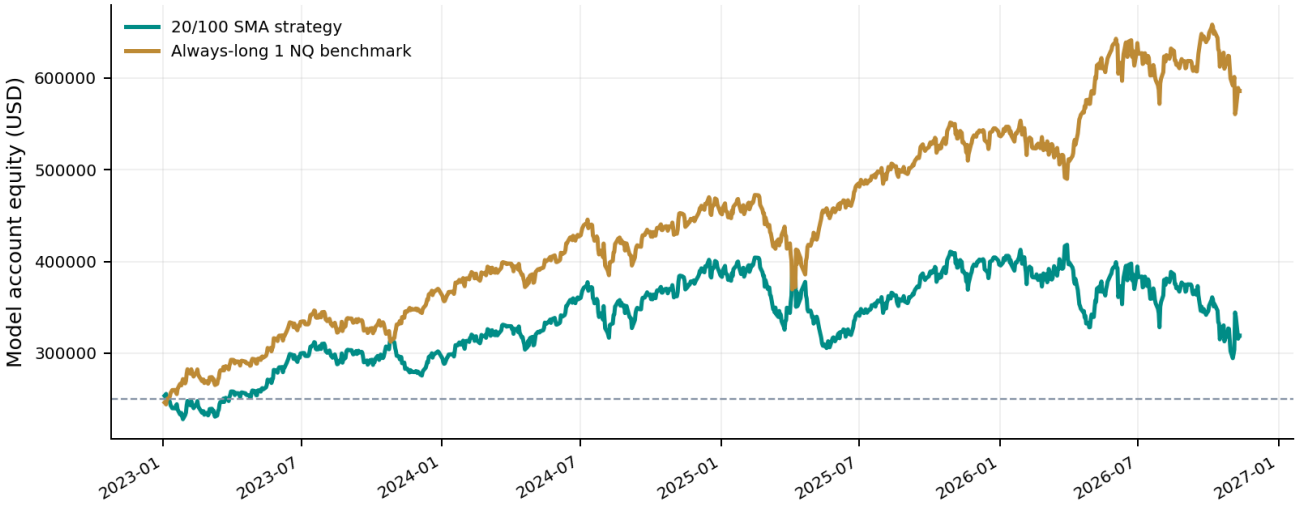


Figure 1. Both curves start at $250,000 on the evaluation window; net simulated P&L includes assumed entry/reversal costs. The always-long futures baseline is leveraged exposure, not the cash Nasdaq-100 index.

Year-by-year comparison

Calendar year	Trend strategy	Always-long NQ	Difference
2023	$49,815	$117,930	-$68,115
2024	$85,000	$85,000	$0
2025	$11,720	$84,190	-$72,470
2026	-$77,385	$48,420	-$125,805

The trend rule matched the long benchmark in 2024 because its exposure remained long throughout that year. Most of the evaluation-period gap developed during periods when the trend rule was short or changed direction while the baseline stayed long. In 2026 through 11 November, the modelled strategy lost $77,385 while the long baseline gained $48,420. This is an attribution observation rather than proof of a causal mechanism.

Trade-level diagnostics

Completed evaluation-period trades: 9; completed-trade win rate: 44.4%. One position remains open at the final recorded open and is excluded from the completed-trade win rate. A nine-trade sample is too small to establish a reliable win probability. The initial training-period trade ledger is kept separately in the CSV output.

VALIDATION / MARKET BEHAVIOUR

Statistical evidence and data-quality audit

Exploratory daily-return diagnostics

Indicator	Result	What this suggests
Mean return after prior up day	0.029%	Descriptive, not a trading signal
Mean return after prior down day	0.165%	Difference can be sample-driven
Welch two-sample p-value	0.0533	Not below the usual 0.05 threshold
Lag-one absolute-return correlation	0.243	Some volatility clustering in sample
Pearson exploratory p-value	5.45e-28	Naive inference; serial dependence matters

The Welch test provides no conventional 5% significance for a return-direction difference. The positive lag-one absolute-return correlation is consistent with volatility clustering; the reported p-value assumes a textbook sampling model and does not establish predictive tradeability. The study includes no multiple-testing or dependence-robust correction.

Historical data screening

Validation measure	Result
Daily records and range	1980 rows; 2019-01-02 - 2026-11-11
Flagged unusually large daily gaps	16
Absolute open-to-open moves above 8%	5
Absolute previous-close-to-open gaps above 2%	12
Bars with at least one off-grid price	30

Flags are screening alerts, not confirmed bad prices. Several of the largest daily movements occur during March 2020 and April 2025; these could reflect volatile markets, contract transitions, data construction or combinations of them. This project does not independently verify which. The numerical results therefore remain indicative and cannot be claimed to be fully roll-safe.

Data representation

The packaged CSV reproduces the public archive's daily Date/Open/High/Low/Close/Volume records, rounding OHLC prices to two decimals. This cent rounding makes it a derived snapshot, not byte-identical source data. The original is available at github.com/Chief-rich/nasdaq100-data/blob/main/data/nq_daily/history.csv. The packaged derived series was cross-checked against the archive to cent precision using a whole-series rolling checksum.

REPRODUCIBILITY / MODEL GOVERNANCE

Execution design, sensitivity and limitations

01 - Return construction and terminal treatment

Each completed date t produces a 20/100 close-based signal. The position is taken no earlier than open(t+1) and earns open-to-open price movement only while that position is held. Changes at an observed open incur one execution side for a flat-to-position change and two sides for a reversal. The final observed open is retained for the closed-trade ledger and execution costs; no unobserved next-open return is fabricated. Any still-open trade is excluded from closed-trade statistics.

02 - Transaction-cost stress test

Cost per side	OOS net P&L	OOS Sharpe
$0.00	$69,300	0.198
$7.50	$69,150	0.198
$15.00	$69,000	0.197
$30.00	$68,700	0.197

Transaction costs modestly affect this low-turnover model, but real market impact, variable slippage, contract-roll trades, margin policy and potential liquidation are not modelled. Cost stress alone does not certify implementability.

03 - What is deliberately not claimed

No exchange-authenticated contract-by-contract prices or verified back-adjustment schedule; no audited execution fills; no interest on collateral, financing, margin calls, taxes or detailed exchange fees; no trade capacity analysis. The 2023+ evaluation is chronological but not a prospectively sealed, out-of-time experiment. Statistical significance can be distorted by serial dependence and model selection.

04 - How to reproduce and independently inspect

Action	Location / command
Install	python -m pip install -r requirements.txt
Test terminal trade accounting	python -m pytest -q
Run archived snapshot	python run.py --csv nq_daily.csv
Review underlying observations	outputs/daily_backtest.csv; outputs/closed_trades.csv
See visual research dashboard	Open_Dashboard.html or outputs/dashboard_results.html
Review flagged dates	outputs/data_quality_flags.csv

05 - Conclusion and next research step

On this specific continuous-futures source, a basic daily 20/100 SMA signal does not outperform simply holding one NQ futures contract long across the 2023-2026 evaluation. A more credible institutional research extension would reconstruct a point-in-time contract chain with an explicit roll policy, margin and collateral rules, then retest on a newly reserved forward period.